

SEMESTER II

Semester	2
Course Name	BIOSAFETY AND BIOETHICS
Course Code	BTN12101
Credits	03
Contact hours/ week	2-1-0 (LTP)
Prerequisite	Nil
Course objective	1. To make students familiar with key concepts of bio-ethics and bio-safety. 2. To understand the various ethical issues involved in research and development to aware about various ethical conflicts.
Course outcome	- Understand basic principles of biosafety and bioethics and its possible impact on biological sciences in both academia and industries - Realize the importance of biosafety practices and guidelines in research activities.

Theory component:**UNIT I: [8L]**

Biosafety: Containment for biohazards; bio-safety levels and assessment; GRAS, GMOs & LMOs.

UNIT II: [10L]

Safety Issues: Laboratory design and facilities, Basic laboratory guidelines, Institutional bioethics and biosafety committees

UNIT III: [8L]

Bioethics: Ethical conflicts and legal issues. Ethics of new technology.

UNIT IV: [5L]

Plagiarism, Conflict of Interest, Ethical Issues in research with living organisms.

Text Books:

- IPR, Biosafety and Bioethics, Goel D & Prashar S – Pearson.
- Industrial documents/ biosafety guidelines

Reference Books

- IPR, Biosafety and biotechnology Management. Senthil Kumar Sand Mohammed Jaabir,-Jasen Publications.
- Biological Safety: Principles and Practices, Dawn P. Wooley and Karen B. Byers, Wiley

	PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
C0								
C01			√		√	√	√	√
C02			√		√	√	√	√